

main.c

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int i, NOP, sum = 0, count = 0, y, quant;
6      int wt = 0, tat = 0;
7      int at[10], bt[10], temp[10];
8      float avg_wt, avg_tat;
9
10     printf("Total number of processes: ");
11     scanf("%d", &NOP);
12
13     y = NOP;
14
15     for(i = 0; i < NOP; i++)
16     {
17         printf("\nEnter Arrival Time and Burst Time for Process[%d]\n", i + 1);
18
19         printf("Arrival Time: ");
20         scanf("%d", &at[i]);
21
22         printf("Burst Time: ");
23         scanf("%d", &bt[i]);
24
25         temp[i] = bt[i]; // store original burst time
26     }
27
28     printf("\nEnter Time Quantum: ");
29     scanf("%d", &quant);
30
31     printf("\nProcess No\tBurst Time\tTAT\tWaiting Time\n");
32
33     for(sum = 0, i = 0; y != 0; )
34     {
35         if(temp[i] <= quant && temp[i] > 0)
36         {
37             sum = sum + temp[i];
38             temp[i] = 0;
39             count = 1;
40         }
41         else if(temp[i] > 0)
42         {
43             temp[i] = temp[i] - quant;
44             sum = sum + quant;
45         }
46
47         if(temp[i] == 0 && count == 1)
48         {
49             y--;
50         }
```

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50
51         printf("P[%d]\\t\\t%d\\t\\t%d\\t\\t%d\\n",
52             i + 1,
53             bt[i],
54             sum - at[i],
55             sum - at[i] - bt[i]);
56
57         wt = wt + (sum - at[i] - bt[i]);
58         tat = tat + (sum - at[i]);
59
60         count = 0;
61     }
62
63     if(i == NOP - 1)
64     {
65         i = 0;
66     }
67     else if(at[i + 1] <= sum)
68     {
69         i++;
70     }
71     else
72     {
73         i = 0;
74     }
75 }
76
77 avg_wt = (float)wt / NOP;
78 avg_tat = (float)tat / NOP;
79
80 printf("\\nAverage Turn Around Time: %.2f", avg_tat);
81 printf("\\nAverage Waiting Time: %.2f\\n", avg_wt);
82
83 return 0;
84 }

```

Total number of process in the system: 3

Enter the Arrival and Burst time of the Process[1]

Arrival time is: 2

Burst time is: 33334

Enter the Arrival and Burst time of the Process[2]

Arrival time is: 23

Burst time is: 45

Enter the Arrival and Burst time of the Process[3]

Arrival time is: 27

Burst time is: 67

Enter the Time Quantum for the process: 9

Process No	Burst Time	TAT	Waiting Time
Process No[2]	45	121	76
Process No[3]	67	175	108
Process No[1]	33334	33444	110
Average Turn Around Time:		98.000000	
Average Waiting Time:		11246.666992	